1 Over-the-Air Computation: Foundations and Techniques

The Over-the-Air Computation (OAC) paradigm takes advantage of the superposition property of the wireless medium to enable simultaneous transmissions from multiple devices to be combined in a way that directly computes a desired function of the transmitted data. This section examines the theoretical foundations and practical techniques of OAC computation, providing an overview of the key concepts and methodologies that underpin the approach to distributed computing in wireless networks.

1.a Theoretical Foundations of OAC

- *Signal Superposition in the Multiple Access Channel (MAC)*: The distinct feature of OAC where computation is handled by harnessing interference via simultaneous transmissions [1], [2].
- *Nomographic Functions*: Defining the class of functions computable via OAC, which can be expressed as a post-processed sum of pre-processed sensor readings [1], [2], [3], [4].
- *Examples of Nomographic Functions*: Arithmetic mean, weighted sum, maximum/minimum value, and modulo-2 sum[1], [2], [5].

1.b OAC Schemes and Implementation Strategies

1.b.i Nomographic Functions

The nomographic functions are the functions that depict the behavior of the OAC schemes. In this thesis project I will focus on how different schemes can be used to compute the arithmetic mean of the transmitted data.

- *Analog vs. Digital OAC*: Reviewing uncoded analog transmission (often optimal for simple Gaussian sensor networks) versus digital schemes designed for modern systems (e.g., using QAM or one-bit quantization)[5], [6].
- *Coherent Schemes (CSI Dependent)*: Discussing methods requiring Channel State Information (CSI), such as Truncated Channel Inversion (TCI) or Zero-Forcing (ZF) coordination, for phase and amplitude alignment [1], [6], [7].
- *Non-Coherent Schemes (CSI Independent)*: Examining techniques robust to synchronization errors, such as those using Orthogonal Signaling (FSK/PPM)[1], [8].

In this section I will demonstrate the Raised Cosine method in order to take advantage of the signal superposition in the Multiple Access Channel (MAC). More specifically I will implement three devices transmitting simultaneously to a fusion center. The transmitted data will be pre-processed using the Raised Cosine method in order to mitigate the intersymbol interference (ISI) that is inherent in digital communication transceivers. The flowgraph of the implementation is shown in Fig. **??**. In it three devices pick a character from a sequence of charachters, unpack it to separate bytes and perform the mapping to the constellation points. The resulting symbols are then passed through a Raised Cosine filter and transmitted simultaneously to the fusion center. The fusion center performs the matched filtering and the necessary post-processing to compute the desired function of the transmitted data.

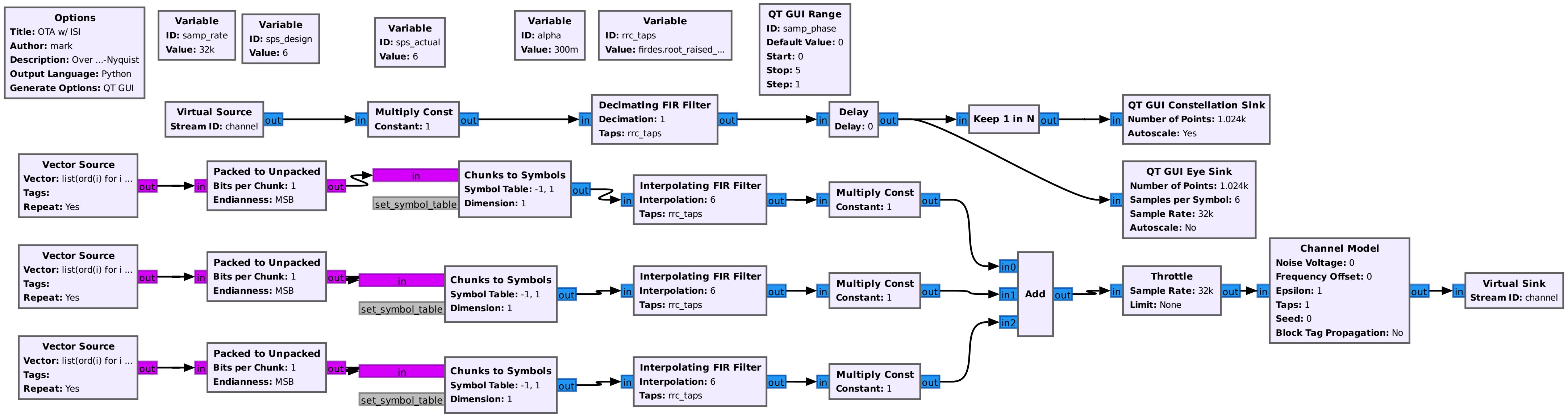

Figure 1: GNU Radio Companion flowgraph for the OTA implementation including ISI considerations.

For the purely analog OAC the RootRaised Cosine can be implemented in GNU Radio using and interpolating filted block at the transmitter and a decimating filter at the receiver. The interpolation/decimation rate is determined by the number of samples per symbol (sps) which is set to 6 in this implementation. The roll-off factor of the filter is set to 0.3, which defines the excess bandwidth of the signal and helps to mitigate intersymbol interference (ISI) while maintaining spectral efficiency. The Root Raised Cosine filter could have been chosen …

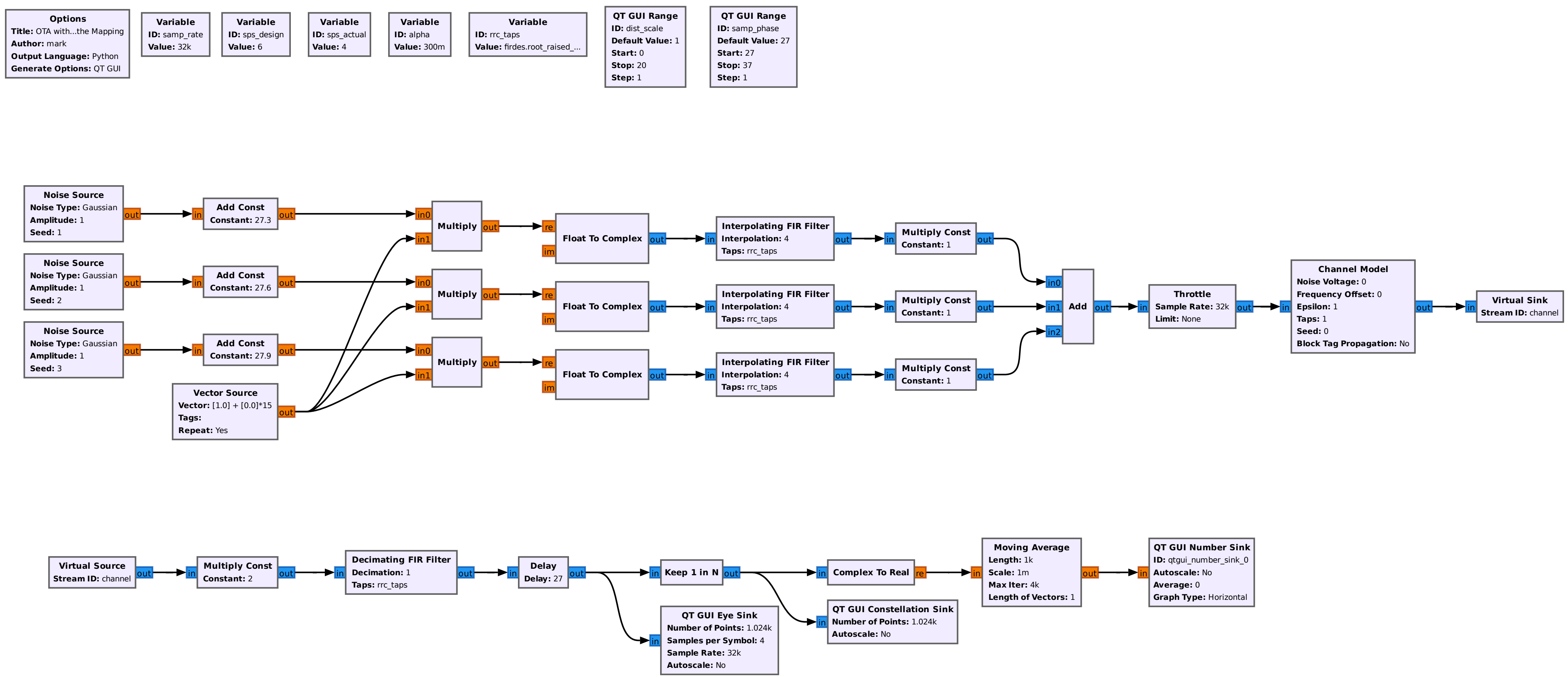

Figure 2: GNU Radio Companion flowgraph for the OTA implementation including ISI considerations.

1.c Applications Context (e.g., Distributed Consensus)

Application choice (e.g., Federated Edge Learning (FEEL)) for gradient aggregation, or average consensus in (Wireless Sensor Networks (WSNs))

[1], [7], [8], [9], [10].

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